

Glaucoma Detection from Fundus Images

A CNN and Transfer Learning Based Classification Study

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Abstract

Glaucoma is a major eye disease that can lead to irreversible vision loss if it is not detected and treated at an early stage. Automated analysis of retinal fundus images using deep learning can assist in identifying images that show signs associated with glaucoma.

In this project, a binary image classification system was developed to classify retinal fundus images into two classes: *glaucoma* and *normal*. A baseline convolutional neural network (CNN) was first developed and evaluated. Class-weighted training was subsequently introduced to address the imbalance between the two classes. Data augmentation was also investigated, although it did not improve the model performance.

Transfer learning using EfficientNetB0 substantially improved the classification performance. The pretrained EfficientNetB0 model was subsequently fine-tuned using a small learning rate and class-weighted training. Finally, the classification threshold was optimized using the validation set to emphasize glaucoma detection.

The final fine-tuned EfficientNetB0 model with a decision threshold of 0.55 achieved an accuracy of 80.88%, glaucoma recall of 68.46%, glaucoma F1-score of 69.04%, macro F1-score of 77.60%, and ROC-AUC of 85.38% on the independent test set.

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1 Introduction

Glaucoma is a progressive eye disease associated with damage to the optic nerve and can result in permanent vision loss. Early detection is therefore important because appropriate intervention can help slow or prevent further progression.

Retinal fundus photography provides a non-invasive way of examining the structures of the eye. Recent advances in deep learning have made convolutional neural networks (CNNs) effective tools for image classification tasks, including medical image analysis.

The objective of this project is to develop a deep learning based binary classifier capable of distinguishing between:

1. Glaucoma
2. Normal

The project follows an experimental approach in which increasingly sophisticated models are compared. The main progression is:

Baseline CNN $\rightarrow$ Class-Weighted CNN $\rightarrow$ Data Augmentation $\rightarrow$ EfficientNetB0 $\rightarrow$
Fine-Tuned EfficientNetB0 $\rightarrow$ Threshold Optimization

A particular emphasis is placed on **glaucoma recall**, since a false negative corresponds to a glaucoma image being classified as normal.

1.1 Project Objectives

The main objectives are:

- Develop a baseline CNN for glaucoma classification.
- Investigate the effect of class-weighted training.
- Investigate the effect of image augmentation.
- Apply transfer learning using EfficientNetB0.
- Fine-tune the pretrained EfficientNetB0 model.
- Optimize the classification threshold using validation data.
- Compare all models using accuracy, precision, recall, F1-score, macro F1-score, and ROC-AUC.
- Analyze the final model using a confusion matrix.

2 Dataset

The dataset used in this project is the *Fondo del Ojo Normal Glaucoma Retinopatía* dataset available on Kaggle.

<https://www.kaggle.com/datasets/jeremypoveda/fondo-del-ojo-normal-glaucoma-retinopatia>

The dataset contains retinal fundus images organized into two classes:

- **glaucoma**
- **normal**

The dataset was divided into separate training and test directories.

2.1 Dataset Distribution

The training dataset contained:

Table 1: Training dataset distribution

Class	Number of Images
Glaucoma	6,950
Normal	10,880
Total	17,830

The test dataset contained:

Table 2: Test dataset distribution

Class	Number of Images
Glaucoma	241
Normal	533
Total	774

The training data were further divided into training and validation subsets using an 80:20 stratified split.

The resulting distributions were:

Table 3: Training and validation distributions

Class	Training	Validation
Glaucoma	5,599	1,351
Normal	8,665	2,215
Total	14,264	3,566

The test set was kept separate and was used for final model evaluation.

3 Data Preprocessing

All images were resized to:

$$224 \times 224 \times 3$$

where the three channels correspond to RGB.

The class labels were encoded as:

$$0 \rightarrow \text{Glaucoma}$$

$$1 \rightarrow \text{Normal}$$

The images were loaded using TensorFlow’s image processing utilities.

For the EfficientNet experiments, the images were resized and converted to floating-point representation. EfficientNetB0’s built-in input preprocessing was used rather than manually applying an additional normalization step.

3.1 Class Imbalance

The dataset contains more normal images than glaucoma images. The training distribution was approximately:

$$\frac{6950}{17830} \approx 39.0\%$$

glaucoma and

$$\frac{10880}{17830} \approx 61.0\%$$

normal images.

Although this is not an extremely severe imbalance, it can cause a classifier to favor the majority class. This was particularly evident in the baseline model.

To address this issue, class-weighted training was investigated.

The approximate class weights used were:

$$w_{\text{glaucoma}} \approx 1.27$$

$$w_{\text{normal}} \approx 0.82$$

Thus, errors involving glaucoma images contributed more strongly to the training loss.

4 Evaluation Metrics

Several evaluation metrics were used.

4.1 Accuracy

Accuracy is defined as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Although useful, accuracy alone is not sufficient for this problem because it does not distinguish between the consequences of different types of errors.

4.2 Precision

For glaucoma:

$$Precision = \frac{TP}{TP + FP}$$

Precision measures the proportion of images predicted as glaucoma that are actually glaucoma.

4.3 Recall

Glaucoma recall is particularly important:

$$Recall = \frac{TP}{TP + FN}$$

A low recall means that many glaucoma cases are incorrectly classified as normal.

4.4 F1-score

The F1-score is the harmonic mean of precision and recall:

$$F1 = 2 \frac{Precision \times Recall}{Precision + Recall}$$

4.5 Macro F1

Macro F1 calculates the F1-score independently for each class and then averages the two values. It therefore gives equal importance to both classes.

4.6 ROC-AUC

The ROC-AUC measures the ability of the model to rank positive and negative samples across different classification thresholds.

Unlike accuracy or F1-score, ROC-AUC is independent of a particular decision threshold.

5 Baseline CNN

A simple convolutional neural network was first implemented as a baseline model.

The architecture consisted of three convolutional blocks followed by fully connected layers:

Input $\rightarrow$ Conv2D $\rightarrow$ MaxPooling $\rightarrow$ Conv2D $\rightarrow$ MaxPooling $\rightarrow$ Conv2D $\rightarrow$
MaxPooling $\rightarrow$ Flatten $\rightarrow$ Dense $\rightarrow$ Dropout $\rightarrow$ Sigmoid

The final sigmoid unit produced the probability corresponding to class 1 (normal).

The baseline model was trained using binary cross-entropy loss and the Adam optimizer.

6 Class-Weighted CNN

The baseline CNN showed a strong tendency to classify images as normal. The test confusion matrix of the baseline model was:

Table 4: Baseline CNN confusion matrix

	Predicted Glaucoma	Predicted Normal
Actual Glaucoma	56	185
Actual Normal	21	512

Only 56 out of 241 glaucoma images were correctly identified.

Therefore, class-weighted training was introduced. The same CNN architecture was retained so that the effect of class weighting could be isolated.

The class-weighted model produced the following test confusion matrix:

Table 5: Class-weighted CNN confusion matrix

	Predicted Glaucoma	Predicted Normal
Actual Glaucoma	81	160
Actual Normal	20	513

Glaucoma recall increased from 23.24% to 33.61%.

7 Data Augmentation

Data augmentation was investigated to determine whether artificially increasing the diversity of the training images could improve generalization.

The augmentation operations included mild transformations such as:

- Horizontal flipping
- Small rotations

- Small zooming
- Small translations
- Mild contrast variation

The augmented model was also trained using class weights.

However, the resulting performance decreased compared with the class-weighted CNN. Therefore, augmentation was not retained as part of the final model.

The experiment was useful because it demonstrated that the selected augmentation strategy did not improve performance on this dataset.

8 Transfer Learning with EfficientNetB0

The next experiment used EfficientNetB0 pretrained on ImageNet.

EfficientNet is a family of convolutional neural networks designed to achieve a favorable balance between accuracy and computational efficiency.

The pretrained convolutional base was initially frozen, and a new classification head was added:

EfficientNetB0 $\rightarrow$ Global Average Pooling $\rightarrow$ Dense(128) $\rightarrow$ Dropout $\rightarrow$ Sigmoid

The pretrained feature extractor was initially kept frozen, and only the newly added classification layers were trained.

Class-weighted training was retained because the earlier experiment demonstrated its usefulness.

8.1 Frozen EfficientNet Results

The frozen EfficientNetB0 model achieved:

Table 6: Frozen EfficientNetB0 test performance

Metric	Score
Accuracy	82.04%
Glaucoma Precision	81.48%
Glaucoma Recall	54.77%
Glaucoma F1	65.51%
Macro Precision	81.84%
Macro Recall	74.57%
Macro F1	76.68%
ROC-AUC	86.32%

This represented a substantial improvement over the CNN-based models.

9 Fine-Tuning EfficientNetB0

Although the frozen EfficientNet model performed well, its convolutional base was originally trained on ImageNet rather than retinal fundus images.

Fine-tuning was therefore performed to allow the higher-level pretrained features to adapt to the target domain.

The earlier layers were kept frozen while approximately the final 30 layers were made trainable. Batch normalization layers were kept frozen to improve training stability.

A small learning rate of:

$$10^{-5}$$

was used during fine-tuning.

Class-weighted training was retained.

9.1 Fine-Tuned Results at Threshold 0.50

At the default threshold of 0.50, the fine-tuned model produced:

Table 7: Fine-tuned EfficientNetB0 at threshold 0.50

Metric	Score
Accuracy	81.40%
Glaucoma Precision	72.15%
Glaucoma Recall	65.56%
Glaucoma F1	68.70%
Macro Precision	78.60%
Macro Recall	77.06%
Macro F1	77.73%
ROC-AUC	85.38%

Fine-tuning increased glaucoma recall from 54.77% to 65.56%.

10 Decision Threshold Optimization

The default classification threshold of 0.50 is not necessarily optimal when the cost of false negatives is important.

In this project, the sigmoid output represents the probability of class 1, which corresponds to the normal class:

$$1 \rightarrow \text{Normal}$$

Therefore, increasing the threshold makes the classifier more likely to assign an image to the glaucoma class.

Thresholds ranging from 0.20 to 0.80 were evaluated on the validation set.

The validation results were:

Table 8: Threshold optimization on validation data

Threshold	Glaucoma Precision	Glaucoma Recall	Glaucoma F1
0.20	0.9481	0.6568	0.7760
0.25	0.9307	0.6957	0.7962
0.30	0.9199	0.7266	0.8119
0.35	0.9038	0.7504	0.8200
0.40	0.8926	0.7712	0.8275
0.45	0.8778	0.7957	0.8347
0.50	0.8642	0.8237	0.8435
0.55	0.8394	0.8496	0.8445
0.60	0.8127	0.8741	0.8423
0.65	0.7837	0.8914	0.8341
0.70	0.7500	0.9129	0.8235
0.75	0.7215	0.9338	0.8140
0.80	0.6784	0.9561	0.7937

A threshold of 0.55 produced the highest validation glaucoma F1-score and was therefore selected without using the test set.

11 Final Model Evaluation

The selected threshold of 0.55 was applied once to the independent test set.

The final confusion matrix was:

Table 9: Final fine-tuned EfficientNetB0 confusion matrix

	Predicted Glaucoma	Predicted Normal
Actual Glaucoma	165	76
Actual Normal	72	461

Therefore:

$$TP = 165, \quad FN = 76, \quad FP = 72, \quad TN = 461$$

The final test performance was:

Table 10: Final model performance on the independent test set

Metric	Score
Decision Threshold	0.55
Accuracy	80.88%
Glaucoma Precision	69.62%
Glaucoma Recall	68.46%
Glaucoma F1	69.04%
Macro Precision	77.73%
Macro Recall	77.48%
Macro F1	77.60%
ROC-AUC	85.38%

The final model correctly identified 165 of the 241 glaucoma images:

$$\frac{165}{241} = 0.6846$$

corresponding to a glaucoma recall of 68.46%.

12 Comparison of Models

The complete model comparison is shown below.

Table 11: Comparison of all models on the test set

Model	Accuracy	G. Precision	G. Recall	G. F1	Macro F1	ROC-AUC
Baseline CNN	73.39%	72.73%	23.24%	35.22%	59.24%	68.64%
Class-Weighted CNN	76.74%	80.20%	33.61%	47.37%	66.22%	74.68%
CNN + Augmentation	71.19%	59.38%	23.65%	33.83%	57.71%	65.75%
EfficientNetB0	82.04%	81.48%	54.77%	65.51%	76.68%	86.32%
Fine-Tuned EfficientNetB0	81.40%	72.15%	65.56%	68.70%	77.73%	85.38%
Fine-Tuned EfficientNetB0 + Threshold 0.55	80.88%	69.62%	68.46%	69.04%	77.60%	85.38%

The results demonstrate that transfer learning provided the largest improvement over the baseline CNN.

The progression in glaucoma recall was:

$$23.24\% \rightarrow 33.61\% \rightarrow 23.65\% \rightarrow 54.77\% \rightarrow 65.56\% \rightarrow \boxed{68.46\%}$$

for the baseline CNN, class-weighted CNN, augmented CNN, EfficientNetB0, fine-tuned EfficientNetB0, and threshold-optimized fine-tuned EfficientNetB0, respectively.

13 Discussion

The experiments provide several important observations.

13.1 Effect of Class Weighting

The baseline CNN exhibited a strong tendency to predict the normal class. Class weighting increased glaucoma recall from 23.24% to 33.61%.

This demonstrates that assigning greater loss weight to the minority glaucoma class can encourage the model to pay greater attention to glaucoma examples.

13.2 Effect of Data Augmentation

The selected augmentation strategy did not improve performance. The augmented CNN achieved a glaucoma recall of only 23.65% and an ROC-AUC of 65.75%.

Therefore, augmentation should not automatically be assumed to improve a medical image classifier. The usefulness of augmentation depends on whether the transformations preserve the relevant visual characteristics of the disease.

13.3 Effect of Transfer Learning

EfficientNetB0 produced a major improvement.

The ROC-AUC increased from:

$$68.64\% \rightarrow 86.32\%$$

when moving from the baseline CNN to EfficientNetB0.

Glaucoma recall also increased from:

$$23.24\% \rightarrow 54.77\%$$

This indicates that the pretrained feature representation was substantially more effective than training a relatively small CNN from scratch.

13.4 Effect of Fine-Tuning

Fine-tuning further improved glaucoma recall:

$$54.77\% \rightarrow 65.56\%$$

at the default threshold.

This suggests that adapting the higher-level pretrained features to retinal fundus images was beneficial.

13.5 Effect of Threshold Optimization

The threshold was selected using validation data rather than the test set. A threshold of 0.55 maximized glaucoma F1 on the validation set.

Applying this threshold to the test set increased glaucoma recall from:

$$65.56\% \rightarrow 68.46\%$$

while glaucoma precision decreased from:

$$72.15\% \rightarrow 69.62\%.$$

This represents the expected precision-recall trade-off associated with changing the classification threshold.

14 Limitations

Several limitations should be considered.

- The dataset is relatively imbalanced, with more normal images than glaucoma images.
- The final test set contains only 241 glaucoma images. Therefore, performance estimates may have uncertainty.
- The model was trained and evaluated on a single dataset. Its generalization to images from different hospitals, cameras, populations, and acquisition protocols has not been established.
- The model is a classification system and does not provide clinical explanations or localization of the retinal structures responsible for its predictions.
- Threshold optimization was performed for glaucoma F1-score. A different threshold may be appropriate if the objective is to maximize sensitivity, specificity, or another clinically defined utility.
- The reported results should not be interpreted as evidence that the model is suitable for clinical diagnosis without extensive external validation and clinical evaluation.

15 Conclusion

This project developed and evaluated a deep learning pipeline for binary classification of retinal fundus images into glaucoma and normal classes.

A simple CNN was first established as a baseline. Its glaucoma recall was only 23.24%, indicating that the baseline model frequently classified glaucoma images as normal.

Class-weighted training improved glaucoma recall to 33.61%. However, the investigated augmentation strategy did not improve performance.

Transfer learning with EfficientNetB0 produced a substantial improvement, achieving an ROC-AUC of 86.32% and glaucoma recall of 54.77%.

Fine-tuning the pretrained EfficientNetB0 further improved glaucoma recall to 65.56%. Finally, the classification threshold was optimized using validation data. A threshold of 0.55 increased the final test glaucoma recall to 68.46% and glaucoma F1-score to 69.04%.

The final model achieved:

Accuracy	= 80.88%
Glaucoma Precision	= 69.62%
Glaucoma Recall	= 68.46%
Glaucoma F1	= 69.04%
Macro F1	= 77.60%
ROC-AUC	= 85.38%

The experimental progression demonstrates the value of transfer learning and domain-specific fine-tuning for retinal image classification. In particular, the substantial improvement in glaucoma recall shows that optimizing the model for the relevant classification objective can provide a more informative assessment than accuracy alone.

16 Future Work

Several extensions could be considered:

- Evaluate other pretrained architectures such as ResNet, DenseNet, or ConvNeXt.
- Use cross-validation to obtain more robust estimates of model performance.
- Evaluate the model on an external glaucoma dataset to assess generalization.
- Investigate explainability techniques such as Grad-CAM to identify regions contributing to model predictions.
- Optimize the threshold according to a specific sensitivity or specificity target rather than F1-score.
- Investigate image-quality assessment and preprocessing methods specifically designed for fundus photography.
- Use ensemble models to investigate whether combining different architectures improves robustness.

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